

Weijie Xu

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RESEARCH INTEREST

Machine Learning, Pattern Recognition, Image Processing
Remote Sensing Applications, Large Language Model Inference, Model-Driven Deep Learning

EDUCATION

Nanjing University of Information Science and Technology

B.E. in Waterford Institute

Nanjing, China

September 2023 - June 2027 (expected)

- GPA: 4.16/5.0
- Rank: 1/34
- Centesimal grade average: 91.45

South East Technological University

B.E. in Computer Science and Information Technology

Munster, Ireland

September 2023 - June 2027 (expected)

- GPA: 4.18/5.0
- Rank: 1/34
- Centesimal grade average: 91.60

AWARDS & HONORS

- **National Scholarship**, 2025
- **Person of the Year**, NUIST, 2025
- **Outstanding Student**, NUIST, 2025
- **Outstanding Student Leader**, NUIST, 2025
- **The Mathematical Contest in Modeling**, Honorable Mention, 2025
- **China Robotics and Artificial Intelligence Competition**, First Prize, 2025
- **National Scholarship**, 2024
- **College Student Innovation and Entrepreneurship Training Program of Jiangsu Province**, 2024
- **National College Student Mathematical Modeling Competition**, First Prize, 2024
- **Higher Mathematics Competition of Jiangsu Province**, First Prize, 2024
- **Outstanding Volunteer**, NUIST, 2024

RESEARCH EXPERIENCE

FSCFNet for Infrared Small Target Detection

Nov 2024 - Sep 2025

- **Role and results:** *IEEE Transactions on Geoscience and Remote Sensing* (Q1 TOP, CCF B), first author, published online.
- Proposed **FSCFNet** for infrared small target detection under sparsity, low contrast, and blurred textures; designed Frequency-Spatial Convolution to jointly model high-frequency details and low-frequency global structures.
- Introduced Asymmetric Cross-Domain Attention and Multiscale Receptive Contextual Block to enhance center responses, multi-scale receptive fields, and long-range dependency modeling.
- On IRSTD-1k, improved detection accuracy from 85.4% to 90.1% (+4.7%) with only +3.6% parameters; achieved leading comprehensive performance against 16 SOTA models on IRSTD-1k, NUAA-SIRST, and NUDT-SIRST.

NVRBench for RGBT Referring Expression Comprehension

Oct 2025 - Present

- **Role and results:** preparing for AAAI (CCF A); Shanghai Jiao Tong University research internship; co-first author.
- Built a semi-automatic benchmark, **NVRBench**, for cross-modal reasoning in drone RGBT referring expression comprehension, targeting missing implicit reasoning annotations and over-reliance on explicit visual descriptions.
- Proposed an implicit-to-explicit chain-of-thought framework (**I2E-CoT**) to guide comparison, verification, and consistency checking under multimodal evidence conflicts.
- Led the end-to-end dataset pipeline, including full manual annotation and quality auditing; conducted systematic literature review and co-authored manuscript drafting and revisions.

TiBA-Fed for Time-Varying Heterogeneity in Edge-Cloud FL
Dec 2025 - Present

- **Role and results:** preparing for *IEEE Transactions on Mobile Computing* (Q1 TOP, CCF A), second author.
- Addressed budget overflow and slow convergence caused by time-varying client heterogeneity by developing a virtual-queue-based drift-plus-penalty optimization framework.
- Jointly optimized client participation and LoRA configuration per round, and designed a heterogeneous local aggregation (HLA) mechanism to mitigate dropout contamination while improving training stability and resource efficiency.
- Contributed to core algorithm design, implemented key modules, reproduced critical experiments (including LoRA configuration and HLA aggregation debugging), and participated in paper writing, revision, and submission.

SKILLS

Programming	Python, C++, Java, SQL, Shell, LaTeX, Pytorch
Operating System, Database	Linux, MySQL, MongoDB, Git, Docker
Software	Office, Obsidian

PUBLICATIONS

- **Think Locally and Act Globally: A Frequency-Spatial Fusion Network for Infrared Small Target Detection**
Weijie Xu[#], Zhenglong Ding^{*}, Ziheng Wang, Zhiqing Cui, Yifan Hu, and Feng Jiang
IEEE Transactions on Geoscience and Remote Sensing, 2025. **IF: 8.6, Rank: Q1 TOP, CCF B.**
- **LESO-Net: A Lightweight and Efficient Small Object Segmentation Network**
Zhenglong Ding[#], Yifan Hu^{*}, Yuanhong Du, **Weijie Xu**, Yamei Wei, and Xuan Yao
Journal of Nanjing University of Information Science and Technology, 2024. **Rank: Core Technology**

OTHER PUBLICATIONS (Under Review)

- **NVRBench: A Cross-Modal Reasoning Benchmark for RGBT Referring Expression Comprehension in Drone Scenes**
Haiyao Jin[#], **Weijie Xu**[#], Siqi Li, Wei Zhang, Zhou Yue, Xue Yang^{*†}, and Junchi Yan[†]
preparing to submit to *AAAI*, 2026. **Rank: CCF A.**
- **TiBA-Fed: Budget-Aware Federated Learning for Time-Varying Heterogeneity Systems in Collaborative Edge-Cloud Intelligence**
Xinkang Fan[#], **Weijie Xu**, Xilei Du, Zirui Xu, and Yi Yao^{*}
submitted to *IEEE Transactions on Mobile Computing* (Under Review), 2026. **IF: 8.1, Rank: Q1 TOP, CCF A.**
- **PFDMNet: Prior-Guided Frequency-Decoupled Mamba Network for Infrared Small Target Detection**
Yifan Hu[#], **Weijie Xu**, Zhenglong Ding^{*}, Kao Zhang, and Feng Jiang
submitted to *Optics and Laser Technology* (in Major Revision), 2026. **IF: 5.7, Rank: Q2.**

SELECTED COURSES

- Linear Algebra (99), College Physics (99), Introduction to Engineering (99), Advanced Mathematics (99), Discrete Mathematics (98), Relational Databases (98), Computer Networks (97), Probability Theory and Statistics (97), Python and its Application Practice (96), Web Application Development (95)